Mars Phoenix Mission Dictionary

NASA Planetary Data System

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USER GUIDE

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The Mars Phoenix mission dictionary (phoenix) contains classes and attributes specific to the Mars Phoenix mission and its instruments.

PDS4 Mars Phoenix Mission Data Dictionary User's Guide
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CHAPTER ONE

INTRODUCTION

1. Purpose of this User's Guide
    - This User's Guide provides an overview of the Mars Phoenix Mission Data Dictionary. The guide details how to include the dictionary in a PDS4 label, describes the organization of the dictionary's classes and attributes, provides definitions for these classes and attributes, and lists example excerpts from labels that use them.
2. Audience
    - This User's Guide should be useful to data providers intending to archive Mars Phoenix data with PDS as well as PDS Nodes who are working with these data providers.

CHAPTER

TWO

OVERVIEW OF THE MARS PHOENIX MISSION DATA DICTIONARY

The Mars Phoenix Mission Data Dictionary contains classes and attributes specific to the Mars Phoenix mission and its instruments.
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CHAPTER THREE

DOCUMENT OUTLINE

1. *How to Include the Mars Phoenix Mission Data Dictionary in a PDS4 Label*
2. *Organization of Classes and Attributes*
    1. *Class Organization*
    2. *Attributes by Class*
3. *Definitions*
    1. *Classes (in alphabetical order)*
    2. *Attributes (in alphabetical order)*
4. *Examples*
5. *Edit History*

CHAPTER

FOUR

HOW TO INCLUDE THE MARS PHOENIX MISSION DATA DICTIONARY IN A PDS4 LABEL

The dictionary consists of a set of files with names in the form PDS4_PHOENIX_xxxx_yyyy.ext, where

- xxxx = the PDS4 Information Model version, e.g. 1Q00
- yyyy = the Mars Phoenix Mission Data Dictionary version, e.g. 1000

and the file extensions are

- .csv = A comma-separated value table of dictionary attributes
- .JSON = The dictionary contents in JSON format
- .sch = The dictionary "rules" as an XML Schematron file
- .txt = The report generated when the dictionary was built
- .xml = The PDS4 label that describes this set of files
- .xsd = The dictionary contents as an XML schema file

Only the schema and Schematron files are needed for validating a PDS4 label.

The latest version of this dictionary may be found on the PDS web site at https://pds.nasa.gov/datastandards/dictionaries/index-missions.shtml#phoenix.

The following is an example showing the use of this dictionary in a PDS4 label.

```
<?xml version="1.0" encoding="UTF-8"?>
<?xml-model href="https://pds.nasa.gov/pds4/pds/v1/PDS4_PDS_1Q00.sch" schematypens=
↪"http://purl.oclc.org/dsdl/schematron"?>
<?xml-model href="https://pds.nasa.gov/pds4/mission/phoenix/v1/PDS4_PHOENIX_1Q00_1000.sch
↪" schematypens="http://purl.oclc.org/dsdl/schematron"?>
<Product_Observational xmlns="http://pds.nasa.gov/pds4/pds/v1"
    xmlns:phoenix="http://pds.nasa.gov/pds4/mission/phoenix/v1"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="https://pds.nasa.gov/pds4/pds/v1/PDS4_PDS_1Q00.xsd
                        https://pds.nasa.gov/pds4/mission/phoenix/v1/PDS4_PHOENIX_1Q00_
↪1000.xsd">
```

The following is a schematic example showing the location of every Mars Phoenix Mission Data Dictionary class and attribute in a PDS4 label. Note that not all classes and attributes may be mutually compatible, and the example does not include any recursion, even if recursion is allowed.

```
<Observation_Area>
  ...
```

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```
  <Mission_Area>
    <phoenix:Phoenix_Parameters>
      <phoenix:Observation_Information>
        <phoenix:mission_phase_name/>
        <phoenix:product_type/>
        <phoenix:product_version_id/>
        <phoenix:spacecraft_clock_start/>
        <phoenix:spacecraft_clock_stop/>
        <phoenix:local_true_solar_time/>
        <phoenix:sol_number/>
        <phoenix:ops_token/>
        <phoenix:ops_token_activity/>
        <phoenix:ops_token_payload/>
        <phoenix:ops_token_command/>
        <phoenix:instument_mode_id/>
      </phoenix:Observation_Information>
    </phoenix:Phoenix_Parameters>
  </Mission_Area>
  ...
</Observation_Area>
```

The namespace for the Mars Phoenix Mission Data Dictionary is http://pds.nasa.gov/pds4/mission/phoenix/v1, abbreviated “phoenix:”.

CHAPTER

FIVE

ORGANIZATION OF CLASSES AND ATTRIBUTES

5.1 Class Organization

Below is a structured list showing the organization of classes, ordered by appearance in the PDS4 label. Each class name is linked to its complete definition in the *Definitions* section.

- *Phoenix_Parameters*
  - *Observation_Information*

5.2 Attributes by Class

The attributes immediately under each class (if any) are listed below. Both classes and attributes are ordered by appearance in the PDS4 label; however, each class is listed only once, even if that class can appear in more than one place in a PDS4 label. Each class and attribute name is linked to its complete definition in the *Definitions* section.

5.2.1 Phoenix_Parameters (attribute list)

5.2.2 Observation_Information (attribute list)

- *mission_phase_name*
- *product_type*
- *product_version_id*
- *spacecraft_clock_start*
- *spacecraft_clock_stop*
- *local_true_solar_time*
- *sol_number*
- *ops_token*
- *ops_token_activity*
- *ops_token_payload*
- *ops_token_command*
- *instument_mode_id*

CHAPTER

SIX

DEFINITIONS

6.1 Classes (in alphabetical order)

6.1.1 Observation_Information

The Observation_Information class provides information about a science observation.

- *go to attribute list*
- Minimum occurrences: 0
- Maximum occurrences: 1

6.1.2 Phoenix_Parameters

The Phoenix_Parameters class is a superclass containing all Phoenix mission classes.

- *go to attribute list*
- Minimum occurrences: 1
- Maximum occurrences: 1

6.2 Attributes (in alphabetical order)

6.2.1 *instument_mode_id*

The instument_mode_id attribute identifies an instrument-dependent designation of operating mode.

- PDS4 data type: ASCII_Short_String_Collapsed
- Valid values:
  - CME
    - Description: Phoenix MECA Instrument Mode = CME, MECA Control and Measurement Electronics.
  - COND
    - Description: Phoenix MECA WCL Instrument Mode = COND, a time-series of electrical conductivity measurements.
  - COUPONS_TABLE

    * Description: Phoenix MECA Instrument Mode = COUPONS_TABLE, one of six onboard tables that modify the operation of the MECA flight software.

  - CP

    * Description: Phoenix MECA WCL Instrument Mode = CP, a time-series of electrode potential and current chronopotentiometry measurements.

  - CV

    * Description: Phoenix MECA WCL Instrument Mode = CV, a time-series of electrode potential and current cyclic voltammetry measurements.

  - DOX

    * Description: Phoenix MECA WCL Instrument Mode = DOX, a time-series of dissolved oxygen measurements acquired with an ion selective electrode with CV-style detection.

  - FRQTEST

    * Description: Phoenix MECA AFM Instrument Mode = FRQTEST, AFM frequency test.

  - ISES

    * Description: Phoenix MECA WCL Instrument Mode = ISES, a time-series of ion selective electrode data.

  - N/A

    * Description: Not applicable.

  - PARAM_VALUE_TABLE

    * Description: Phoenix MECA Instrument Mode = PARAM_VALUE_TABLE, one of six onboard tables that modify the operation of the MECA flight software.

  - PT

    * Description: Phoenix MECA WCL Instrument Mode = PT, a time-series of pressure and temperature measurements.

  - PWR

    * Description: Phoenix MECA Instrument Mode = PWR, a time series of voltage and current values provided by the primary MECA 5V (load and logic) and 15V (load and AFM) power supplies.

  - SCAN

    * Description: Phoenix MECA TECP or AFM Instrument Mode = SCAN, indicating scanning mode.

  - STATE_TABLE

    * Description: Phoenix MECA Instrument Mode = STATE_TABLE, one of six onboard tables that modify the operation of the MECA flight software.

  - TIPS_TABLE

    * Description: Phoenix MECA AFM Instrument Mode = TIPS_TABLE, data resulting from initialization and test of the Wheatstone Bridge used to monitor the deflection of the AFM cantilevers. Or, Phoenix MECA Instrument Mode = TIPS_TABLE, one of six onboard tables that modify the operation of the MECA flight software.

- Minimum Length: 1
- Maximum Length: 255
- Nillable: No

- Minimum occurrences: 0
- Maximum occurrences: 1

6.2.2 *local_true_solar_time*

The local_true_solar_time attribute is the local true solar time, as defined in the main PDS4 data dictionary, at the beginning of an observation.

- PDS4 data type: ASCII_Short_String_Collapsed
- Valid values: N/A
- Minimum Length: 1
- Maximum Length: 255
- Nillable: No
- Minimum occurrences: 0
- Maximum occurrences: 1

6.2.3 *mission_phase_name*

The mission_phase_name identifies a time period within the mission.

- PDS4 data type: ASCII_Short_String_Preserved
- Valid values:
  - EXTENDED MISSION
    - Description: The Mars Phoenix Extended Mission Phase.
  - PRIMARY MISSION
    - Description: The Mars Phoenix Primary Mission Phase.
- Minimum Length: 1
- Maximum Length: 255
- Nillable: No
- Minimum occurrences: 0
- Maximum occurrences: 1

6.2.4 *ops_token*

The ops_token is a code associated with a Phoenix data product that provides information about the command sequence that caused the product to be acquired. The value is a 32-bit unsigned hexadecimal integer expressed in a PDS label as 16#AAAAPCCC#, where: - AAAA is the campaign ID assigned by the sequence planning team, - P is a 4-bit payload ID reserved for use by each instrument team, as defined by the team, and - CCC is the command sequence number, which is set to zero for each new campaign and automatically incremented with each command. The combination of ops_token and sol number should uniquely identify a Phoenix data product.

- PDS4 data type: ASCII_Short_String_Collapsed
- Valid values: N/A

- Minimum Length: 1
- Maximum Length: 255
- Nillable: No
- Minimum occurrences: 0
- Maximum occurrences: 1

6.2.5 *ops_token_activity*

ops_token_activity is the 16-bit activity code (campaign id) from the ops_token, represented as a 4-digit Hex value.

- PDS4 data type: ASCII_Short_String_Collapsed
- Valid values: N/A
- Minimum Length: 1
- Maximum Length: 255
- Nillable: No
- Minimum occurrences: 0
- Maximum occurrences: 1

6.2.6 *ops_token_command*

ops_token_command is the 12-bit command sequence number of the ops token, represented as a 3-digit Hex value.

- PDS4 data type: ASCII_Short_String_Collapsed
- Valid values: N/A
- Minimum Length: 1
- Maximum Length: 255
- Nillable: No
- Minimum occurrences: 0
- Maximum occurrences: 1

6.2.7 *ops_token_payload*

ops_token_payload is the 4-bit payload id value of the ops token, represented as a single Hex digit.

- PDS4 data type: ASCII_Short_String_Collapsed
- Valid values: N/A
- Minimum Length: 1
- Maximum Length: 255
- Nillable: No
- Minimum occurrences: 0
- Maximum occurrences: 1

6.2.8 *product_type*

The product_type attribute identifies a group of data products within a collection that have some property in common.

- PDS4 data type: ASCII_Short_String_Collapsed
- Valid values:
  - EGA_AVG_CALLS
    - Description: TEGA Engineering data, Evolved Gas Analyzer Average # of Calls to Task Queue.
  - EGA_AVG_IDLE_CALLS
    - Description: TEGA Engineering data, Evolved Gas Analyzer Average # of Idle Calls to Task Queue.
  - EGA_EMISSION_CUR
    - Description: TEGA Engineering data, Evolved Gas Analyzer Emission Current Monitor.
  - EGA_FILAMENT_1
    - Description: TEGA Engineering data, Evolved Gas Analyzer Filament 1 in Use.
  - EGA_FILAMENT_1_SEL
    - Description: TEGA Engineering data, Evolved Gas Analyzer Filament 1 in Use.
  - EGA_FILAMENT_2
    - Description: TEGA Engineering data, Evolved Gas Analyzer Filament 2 in Use.
  - EGA_FILAMENT_2_SEL
    - Description: TEGA Engineering data, Evolved Gas Analyzer Filament 2 in Use.
  - EGA_FILAMENT_CUR_1
    - Description: TEGA Engineering data, Evolved Gas Analyzer Filament 1 Current Monitor.
  - EGA_FILAMENT_CUR_2
    - Description: TEGA Engineering data, Evolved Gas Analyzer Filament 2 Current Monitor.
  - EGA_GEC_CUR
    - Description: TEGA Engineering data, Evolved Gas Analyzer Gas Enrichment Cell Current Monitor.
  - EGA_ION_PUMP_CUR
    - Description: TEGA Engineering data, Evolved Gas Analyzer Ion Pump Current Monitor.
  - EGA_ION_PUMP_VOLT
    - Description: TEGA Engineering data, Evolved Gas Analyzer Ion Pump Voltage Monitor.
  - EGA_MAGNET_TEMP_1
    - Description: TEGA Engineering data, Evolved Gas Analyzer Magnet 1 Temperature.
  - EGA_MAGNET_TEMP_2
    - Description: TEGA Engineering data, Evolved Gas Analyzer Magnet 2 Temperature.
  - EGA_MINUS_12_VOLT
    - Description: TEGA Engineering data, Evolved Gas Analyzer -12 Monitor.
  - EGA_MIN_CALLS
    - Description: TEGA Engineering data, Evolved Gas Analyzer Minimum # of Calls to Task Queue.

- EGA_MIN_IDLE_CALLS
  * Description: TEGA Engineering data, Evolved Gas Analyzer Minimum # of Idle Calls to Task Queue.
- EGA_MULTIPLIER_VOLT
  * Description: TEGA Engineering data, Evolved Gas Analyzer Multiplier Voltage Monitor.
- EGA_PLUS_12_VOLT
  * Description: TEGA Engineering data, Evolved Gas Analyzer +12 Monitor.
- EGA_PLUS_5_VOLT
  * Description: TEGA Engineering data, Evolved Gas Analyzer +5 Monitor.
- EGA_PROC_TEMP
  * Description: TEGA Engineering data, Evolved Gas Analyzer CPU Temperature.
- EGA_STATUS_BITS
  * Description: TEGA Engineering data, Evolved Gas Analyzer Status Bits Value.
- EGA_SWEEP_VOLT
  * Description: TEGA Engineering data, Evolved Gas Analyzer Sweep Voltage Monitor.
- EGA_SWEEP_VOLTAGE
  * Description: TEGA Engineering data, Evolved Gas Analyzer Sweep Voltage Monitor.
- EGA_TRAP_CUR
  * Description: TEGA Engineering data, Evolved Gas Analyzer Trap Current Monitor.
- MECA-EM0
  * Description: MECA EDR data type 0. AFM: FRQTEST
- MECA-EM1
  * Description: MECA EDR data type 1. AFM: RESPONSE
- MECA-EM10
  * Description: MECA EDR data type 10. WCL: DOX
- MECA-EM11
  * Description: MECA EDR data type 11. WCL: CV
- MECA-EM12
  * Description: MECA EDR data type 12. WCL: CP
- MECA-EM13
  * Description: MECA EDR data type 13. WCL: AS
- MECA-EM14
  * Description: MECA EDR data type 14. WCL: (Reserved)
- MECA-EM15
  * Description: MECA EDR data type 15. WCL: PT
- MECA-EM2
  * Description: MECA EDR data type 2. AFM: SCAN

- MECA-EM3
    * Description: MECA EDR data type 3. AFM: TIPS
- MECA-EM4
    * Description: MECA EDR data type 4. CME_STATUS
- MECA-EM5
    * Description: MECA EDR data type 5. POWER_DATA
- MECA-EM6
    * Description: MECA EDR data type 6. TBL
- MECA-EM7
    * Description: MECA EDR data type 7. TECP
- MECA-EM8
    * Description: MECA EDR data type 8. WCL: ISES
- MECA-EM9
    * Description: MECA EDR data type 9. WCL: COND
- MECA_AFM_REPORT
    * Description: The MECA AFM Reports are text files that describes the measurement day's events.
- MECA_AFM_SDD
    * Description: The MECA AFM Scan derivative records (SDD) are line-by-line first order spatial derivatives of the SDRs, processed using the Savitzky-Golay filter method.
- MECA_AFM_SDR
    * Description: The MECA AFM Scan Data Records (SDR) are AFM scan data that has been converted from DN to physical units.
- MECA_TECP_EC
    * Description: MECA TECP time-series electrical conductivity (EC) data.
- MECA_TECP_HUM
    * Description: TECP time-series frost point data.
- MECA_TECP_PRM
    * Description: MECA TECP time-series relative permittivity (dielectric constant) data.
- MECA_TECP_TC
    * Description: MECA TECP time_series temperature data.
- MECA_WCL_COND
    * Description: MECA WCL time-series of conductivity (COND) data.
- MECA_WCL_CP
    * Description: MECA WCL time-series of electrode potential and current chronopotentiometry (CP) data.
- MECA_WCL_CV
    * Description: MECA WCL time-series of electrode potential and current cyclic voltammetry (CV) data.

- MECA_WCL_ISE
  * Description: MECA WCL time-series of ion selective electrode (ISE) data.
- MECA_WCL_PT
  * Description: MECA WCL time-series of pressure and temperature (PT) data.
- MEM_OVEN_CUR
  * Description: TEGA Engineering data, Thermal Analyzer Oven Current.
- MEM_OVEN_ERR
  * Description: TEGA Engineering data, Thermal Analyzer Oven Error.
- MEM_OVEN_INT_HI
  * Description: TEGA Engineering data, Thermal Analyzer Oven Integrator Value High Bytes.
- MEM_OVEN_INT_LO
  * Description: TEGA Engineering data, Thermal Analyzer Oven Integrator Value Low Bytes.
- MEM_OVEN_VOLT
  * Description: TEGA Engineering data, Thermal Analyzer Oven Voltage.
- MEM_OVEN_WIDTH
  * Description: TEGA Engineering data, Thermal Analyzer Oven Pulse Width.
- MEM_SHLD_CUR
  * Description: TEGA Engineering data, Thermal Analyzer Shield Current.
- MEM_SHLD_ERR
  * Description: TEGA Engineering data, Thermal Analyzer Shield Error.
- MEM_SHLD_INT_HI
  * Description: TEGA Engineering data, Thermal Analyzer Shield Integrator Value High Bytes.
- MEM_SHLD_INT_LO
  * Description: TEGA Engineering data, Thermal Analyzer Shield Integrator Value Low Bytes.
- MEM_SHLD_VOLT
  * Description: TEGA Engineering data, Thermal Analyzer Shield Voltage.
- MEM_SHLD_WIDTH
  * Description: TEGA Engineering data, Thermal Analyzer Shield Pulse Width.
- MEM_T_WIDTH
  * Description: TEGA Engineering data, Thermal Analyzer T Heater Pulse Width.
- TA_A2D_TEMP
  * Description: TEGA Engineering data, Thermal Analyzer Analog to Digital Converter Temperature.
- TA_AGD_0_3
  * Description: TEGA Engineering data, Thermal Analyzer AGD_0_3 Ground.
- TA_AGD_3_1
  * Description: TEGA Engineering data, Thermal Analyzer AGD_3_1 Ground.

- TA_ANLG_MINUS_12_CUR
  * Description: TEGA Engineering data, Thermal Analyzer Analog -12 Current.
- TA_ANLG_MINUS_12_VOLT
  * Description: TEGA Engineering data, Thermal Analyzer Analog -12 Voltage.
- TA_ANLG_PLUS_12_CUR
  * Description: TEGA Engineering data, Thermal Analyzer Analog +12 Current.
- TA_ANLG_PLUS_12_VOLT
  * Description: TEGA Engineering data, Thermal Analyzer Analog +12 Voltage.
- TA_BUS_A_CUR
  * Description: TEGA Engineering data, Thermal Analyzer Bus A Current.
- TA_BUS_A_VOLT
  * Description: TEGA Engineering data, Thermal Analyzer Bus A Voltage.
- TA_BUS_B_CUR
  * Description: TEGA Engineering data, Thermal Analyzer Bus B Current.
- TA_CAL_TANK_COLD_TEMP
  * Description: TEGA Engineering data, Thermal Analyzer Cal Tank Cold Temperature.
- TA_CAL_TANK_TEMP
  * Description: TEGA Engineering data, Thermal Analyzer Cal Tank Temperature.
- TA_COVER_1_TEMP
  * Description: TEGA Engineering data, Thermal Analyzer Cover 1 Temperature.
- TA_COVER_2_TEMP
  * Description: TEGA Engineering data, Thermal Analyzer Cover 2 Temperature.
- TA_CPU_PLUS_5_CUR
  * Description: TEGA Engineering data, Thermal Analyzer CPU +5 Current.
- TA_CPU_PLUS_5_VOLT
  * Description: TEGA Engineering data, Thermal Analyzer CPU +5 Voltage.
- TA_CPU_TEMP
  * Description: TEGA Engineering data, Thermal Analyzer CPU Temperature.
- TA_EGA_BAKEOUT_TEMP
  * Description: TEGA Engineering data, Thermal Analyzer Evolved Gas Analyzer Bakeout Temperature.
- TA_EGA_CUR
  * Description: TEGA Engineering data, Thermal Analyzer Evolved Gas Analyzer Current.
- TA_EGA_ELECT_BOX_TEMP
  * Description: TEGA Engineering data, Thermal Analyzer Evolved Gas Analyzer Electronics Temperature.
- TA_EGA_GEC_TEMP

    * Description: TEGA Engineering data, Thermal Analyzer Evolved Gas Analyzer Gas Enrichment Cell Temperature.

- TA_EGA_MAN_TEMP
    * Description: TEGA Engineering data, Thermal Analyzer Evolved Gas Analyzer Manifold Temperature.
- TA_EGA_PLUMB_TEMP
    * Description: TEGA Engineering data, Thermal Analyzer Evolved Gas Analyzer Plumbing Temperature.
- TA_FULL_DETECT
    * Description: TEGA Engineering data, Thermal Analyzer Full Detect Integrated Diode Sensor Reading.
- TA_FULL_DETECT_RAW
    * Description: TEGA Engineering data, Thermal Analyzer Full Detect Raw Diode Sensor Reading.
- TA_INPUT_FUNNEL_1_LO_TEMP
    * Description: TEGA Engineering data, Thermal Analyzer Input Funnel 1 Low Temperature.
- TA_INPUT_FUNNEL_2_LO_TEMP
    * Description: TEGA Engineering data, Thermal Analyzer Input Funnel 2 Low Temperature.
- TA_MANIFOLD_PRES
    * Description: TEGA Engineering data, Thermal Analyzer Manifold Pressure.
- TA_MANIFOLD_TEMP
    * Description: TEGA Engineering data, Thermal Analyzer Manifold Temperature.
- TA_OUTLET_PRES
    * Description: TEGA Engineering data, Thermal Analyzer Outlet Pressure.
- TA_OVEN_ERR
    * Description: TEGA Engineering data, Thermal Analyzer Oven Error.
- TA_OVEN_PLUS_15_CUR
    * Description: TEGA Engineering data, Thermal Analyzer Oven +15 Current.
- TA_OVEN_PLUS_15_VOLT
    * Description: TEGA Engineering data, Thermal Analyzer Oven +15 Voltage.
- TA_OVEN_TEMP
    * Description: TEGA Engineering data, Thermal Analyzer Oven Temperature.
- TA_PLUMBING_1_TEMP
    * Description: TEGA Engineering data, Thermal Analyzer Plumbing 1 Temperature.
- TA_PLUMBING_2_TEMP
    * Description: TEGA Engineering data, Thermal Analyzer Plumbing 2 Temperature.
- TA_PLUS_5_VREF
    * Description: TEGA Engineering data, Thermal Analyzer Plus 5 Voltage Reference.

- TA_PRES_SENSE_FD_BK
  - Description: TEGA Engineering data, Thermal Analyzer Pressure Sensor Exc. Feedback.
- TA_PWR_CNTL_1_TEMP
  - Description: TEGA Engineering data, Thermal Analyzer Power Control 1 Temperature.
- TA_PWR_CNTL_2_TEMP
  - Description: TEGA Engineering data, Thermal Analyzer Power Control 2 Temperature.
- TA_PWR_SPLY_1_TEMP
  - Description: TEGA Engineering data, Thermal Analyzer Power Supply 1 Temperature.
- TA_PWR_SPLY_2_TEMP
  - Description: TEGA Engineering data, Thermal Analyzer Power Supply 2 Temperature.
- TA_SHIELD_PLUS_30_CUR
  - Description: TEGA Engineering data, Thermal Analyzer Shield +30 Current.
- TA_SHIELD_PLUS_30_VOLT
  - Description: TEGA Engineering data, Thermal Analyzer Shield +30 Voltage.
- TA_SHLD_ERR
  - Description: TEGA Engineering data, Thermal Analyzer Shield Error.
- TA_SHLD_TEMP
  - Description: TEGA Engineering data, Thermal Analyzer Shield Temperature.
- TA_TRANS_TUBE_TEMP
  - Description: TEGA Engineering data, Thermal Analyzer Transfer Tube Temperature.
- TA_T_HEATER_TEMP
  - Description: TEGA Engineering data, Thermal Analyzer T Heater Temperature.
- TEGA_EGAEDR
  - Description: TEGA EGA EDR (EGAEDR) data.
- TEGA_EGHEDR
  - Description: TEGA EGA Mass Hopping Counts EDR (EGHEDR) data.
- TEGA_EGHRDR
  - Description: TEGA EGA Mass Hopping Mode RDR (EGHRDR) data.
- TEGA_EGS
  - Description: TEGA EGA Sweep Mode (EGS) RDR data.
- TEGA_E_KERNEL
  - Description: TEGA E-Kernel Report data.
- TEGA_LEDEDR
  - Description: TEGA LED Sensor (LEDEDR) Oven Fill data.
- TEGA_MSGEDR
  - Description: TEGA Message Log EDR (MSGEDR) data.

  - TEGA_SC
    * Description: TEGA Scanning Calorimeter RDR (SCRDR) data.
  - TEGA_SCEDR
    * Description: TEGA Scanning Calorimeter Oven Heating EDR (SCEDR) data.
- Minimum Length: 1
- Maximum Length: 255
- Nillable: No
- Minimum occurrences: 0
- Maximum occurrences: 1

6.2.9 *product_version_id*

The product_version_id attribute identifies the version of an individual data product.

- PDS4 data type: ASCII_Short_String_Collapsed
- Valid values: N/A
- Minimum Length: 1
- Maximum Length: 255
- Nillable: No
- Minimum occurrences: 0
- Maximum occurrences: 1

6.2.10 *sol_number*

The sol_number attibute is the number of the Mars day on which an observation was acquired. Landing day is Sol 0.

- PDS4 data type: ASCII_Short_String_Collapsed
- Valid values: N/A
- Minimum Length: 1
- Maximum Length: 255
- Nillable: No
- Minimum occurrences: 0
- Maximum occurrences: 1

6.2.11 *spacecraft_clock_start*

The spacecraft_clock_start attribute provides the value of the spacecraft clock at the beginning of a time period of interest.

- PDS4 data type: ASCII_Short_String_Collapsed
- Valid values: N/A
- Minimum Length: 1
- Maximum Length: 255
- Nillable: No
- Minimum occurrences: 0
- Maximum occurrences: 1

6.2.12 *spacecraft_clock_stop*

The spacecraft_clock_stop attribute provides the value of the spacecraft clock at the end of a time period of interest.

- PDS4 data type: ASCII_Short_String_Collapsed
- Valid values: N/A
- Minimum Length: 1
- Maximum Length: 255
- Nillable: No
- Minimum occurrences: 0
- Maximum occurrences: 1

CHAPTER

SEVEN

EXAMPLES

Example PDS4 label snippet from urn:nasa:pds:phx_meca_raw:data_afm:fs004em2_00_1080011040000j1::1.0:

```
<Mission_Area>
  <phoenix:Phoenix_Parameters>
    <phoenix:Observation_Information>
      <phoenix:mission_phase_name>PRIMARY MISSION</phoenix:mission_phase_name>
      <phoenix:product_type>MECA-EM2</phoenix:product_type>
      <phoenix:product_version_id>V1.0 D-22850</phoenix:product_version_id>
      <phoenix:spacecraft_clock_start>896567774.000</phoenix:spacecraft_clock_start>
      <phoenix:spacecraft_clock_stop>896567777.000</phoenix:spacecraft_clock_stop>
      <phoenix:local_true_solar_time>12:58:38</phoenix:local_true_solar_time>
      <phoenix:sol_number>4</phoenix:sol_number>
      <phoenix:ops_token>16#11040000#</phoenix:ops_token>
      <phoenix:ops_token_activity>16#00001104#</phoenix:ops_token_activity>
      <phoenix:ops_token_payload>16#00000000#</phoenix:ops_token_payload>
      <phoenix:ops_token_command>16#00000000#</phoenix:ops_token_command>
      <phoenix:instument_mode_id>SCAN</phoenix:instument_mode_id>
    </phoenix:Observation_Information>
  </phoenix:Phoenix_Parameters>
</Mission_Area>
```

CHAPTER

EIGHT

EDIT HISTORY

See also: *PHOENIX change log*.
2026-06-11 Jennifer Ward